

TABLE OF CONTENTS

S L. No	Topic	Page No
1.	Project Synopsis	7-9
1.1	Purpose	7
1.2	Goals	9-10
1.3	Intended Audience	10
2.	Project Objectives	11-12
3.	Project Outline	13-15
3.1	Authentication Module	13
3.2	Profile Management Module	13
3.3	Resume Upload and Analysis Module	13
3.4	Role Selection Module	13
3.5	Dashboard Module	14
3.6	Session Setup Module	14
3.7	Face Verification Module	14
3.8	Question Generation Module	14
3.9	Voice Answer Module	14

3.10	AI Feedback Module	14
3.11	Mock Interview Monitoring Module	15
3.12	Session Summary Module	15
3.13	Session Details Module	15
3.14	Progress Tracking Module	15
3.15	Data Storage Module	15
3.16	AI Integration Module	15
3.17	Flow Diagram	16-18
4.	Technologies Used	19-20
5.	Screenshots	21-25
6.	Project Features	26-28
7.	Test Cases	29-32
8.	User Guide	33-35
9.	Conclusion	36
10.	References	37-38

1. PROJECT SYNOPSIS

The AI Interview Trainer is an Android-based mobile application developed to help students and job seekers prepare for interviews using Artificial Intelligence. The application provides a smart interview practice environment where users can upload their resume, select job roles, attend mock interviews, answer questions using voice, and receive AI-based feedback and performance analysis. The system uses Artificial Intelligence to analyze the user's resume and suggest suitable job roles based on skills and experience. The application then generates interview questions based on the selected role, difficulty level, and interview category such as technical, behavioral, or mixed. The user can answer the questions using voice, and the system converts speech to text and evaluates the answer using AI. The application also includes a mock interview mode where face verification and monitoring are performed to simulate a real interview environment. The system detects if no face or multiple faces are present and shows warnings to maintain interview integrity. After the interview is completed, the system provides a score, feedback, strengths, improvement suggestions, and model answers. The application stores all interview session data and shows progress analytics such as average score, best score, session history, and role-wise performance. This helps the user track improvement over time. The AI Interview Trainer helps users improve their interview skills, confidence, communication, and technical knowledge by providing a real-time AI-powered interview practice platform.

1.1 Purpose

The purpose of the AI Interview Trainer project is to develop an intelligent mobile application that helps students and job seekers prepare for interviews using Artificial Intelligence. The application provides a realistic interview practice environment where users can attend mock interviews, answer questions using voice, and receive AI-based feedback on their performance.

The system is designed to simulate real interview conditions by generating role-based interview questions, evaluating answers, monitoring the user during mock interviews, and tracking performance over time. The application helps users identify their strengths and weaknesses and improve their interview skills through continuous practice and feedback.

The main purpose of this project is to make interview preparation smarter, more efficient, and accessible to everyone using AI technology.

Aims of the Project:

- To provide a secure, admin-only mobile application for managing gym operations.
- To design and develop an AI-powered mobile application for interview preparation.
- To implement resume analysis to identify user skills and suggest suitable job roles.
- To generate interview questions based on selected job role, difficulty level, and interview category.
- To allow users to answer interview questions using voice and convert speech to text.
- To evaluate user answers using Artificial Intelligence and provide feedback and scores.
- To implement mock interview monitoring using face detection to simulate real interview conditions.
- To store interview session details and maintain performance records.
- To display performance analytics such as average score, best score, and progress history.
- To help users improve communication skills, technical knowledge, and interview confidence.
- To create a user-friendly and interactive mobile application for students and job seekers.

1.2 Goals

1. Improve Interview Preparation Using AI

Develop an AI-powered mobile application that helps students and job seekers prepare for interviews by providing role-based interview questions, AI evaluation, and performance feedback in a smart and efficient way.

2. Enable Resume-Based Role Recommendation

Implement a resume analysis system that extracts skills and experience from the uploaded resume and suggests suitable job roles for interview practice.

3. Provide Real-Time Interview Practice Environment

Create a mock interview environment where users can attend interviews, answer questions using voice, and experience a real interview-like scenario.

4. Implement Voice-Based Answer Evaluation

Allow users to answer interview questions using voice recording, convert speech to text, and evaluate the answer using Artificial Intelligence.

5. Monitor User During Mock Interview

Implement face detection and monitoring during mock interviews to ensure the presence of only one candidate and simulate real interview conditions.

6. Track User Performance and Progress

Store interview session data and display performance analytics such as average score, best score, session history, and role-wise performance.

7. Provide AI-Based Feedback and Improvement Suggestions

Generate AI-based feedback including strengths, weaknesses, improvement suggestions, and model answers after each interview session.

8. Store and Manage Interview Data Securely

Use Firebase Authentication and Firebase Firestore to securely store user data, resume details, interview sessions, and performance records.

9. Create a User-Friendly Mobile Application

Design a simple, interactive, and easy-to-use mobile interface so that users can practice interviews without technical difficulty.

10. Help Users Build Confidence and Communication Skills

Provide continuous interview practice and feedback to help users improve confidence, communication skills, and technical knowledge.

1.3 Intended Audience

1. Students Preparing for Placements: Final year students and fresh graduates who are preparing for campus placements and job interviews can use this application to practice interviews, improve communication skills, and gain confidence through AI-based mock interviews and feedback.

2. Job Seekers : Individuals who are searching for jobs and want to improve their interview performance can use this application to practice technical and HR interview questions based on their selected job roles and difficulty levels.

3. Training Institutions and Placement Training Centers : Training institutes and placement preparation centers can use this application as a training tool to help students practice interviews and evaluate their performance using Artificial Intelligence.

4. Academic Evaluators : Faculty members and project evaluators reviewing this project as part of the Mobile Application Development coursework under the BCA programme can use this application to understand the implementation of AI integration, Firebase database, and Android application development.

2. PROJECT OBJECTIVES

The following objectives define the specific goals that AI Interview Trainer aims to achieve through its design and implementation:

2.1 To Develop a Secure User Authentication System

Implement Firebase Authentication to allow users to securely register and log in to the application using email/password or Google Sign-In. The system ensures secure access, user data privacy, and session management so that only authenticated users can access the application features.

2.2 To Implement Profile Management System

Develop a profile management module where users can enter and update personal details such as name, college, graduation year, phone number, and profile photo. This information is stored securely and used for interview session records and performance tracking.

2.3 To Implement Resume Upload and AI-Based Resume Analysis

Allow users to upload their resume in PDF format and analyze it using Artificial Intelligence to extract skills, experience level, and suggest suitable job roles based on the resume content.

2.4 To Implement Role-Based Interview Question Generation

Develop a system that generates interview questions based on the selected job role, difficulty level, and interview category such as technical, behavioral, or mixed using Artificial Intelligence.

2.5 To Implement Voice-Based Answer Recording and Evaluation

Allow users to answer interview questions using voice recording. The system converts speech to text and evaluates the answer using AI to generate scores, feedback, and model answers.

2.6 To Implement Mock Interview Monitoring Using Face Detection

Use face detection to verify the presence of the user during mock interview sessions. The system detects no face or multiple faces and shows warnings to simulate real interview conditions and maintain interview integrity.

2.7 To Store Interview Session Data and Performance Records

Store interview session details such as selected role, questions, answers, scores, and feedback in Firebase Firestore so that users can view their session history and performance records later.

2.8 To Display Performance Analytics and Progress Tracking

Develop a dashboard that displays performance analytics such as total sessions, average score, best score, session history, and role-wise performance to help users track their improvement over time.

2.9 To Ensure a Clean and User-Friendly Interface

Design the application with a simple, clean, and user-friendly interface so that users can easily navigate through the application and use all features without difficulty.

2.10 To Support Scalability and Future Enhancements

Structure the application in a modular architecture so that future features such as video interview analysis, facial expression analysis, and advanced AI feedback can be added easily.

3. PROJECT OUTLINE

The AI Interview Trainer is structured as a multi-module Android application where each module handles a specific function related to interview preparation and performance analysis. The application follows a user-based architecture where each registered user can log in, upload resume, attend mock interviews, and track their performance. The system integrates Artificial Intelligence for resume analysis, question generation, answer evaluation, and performance feedback.

3.1 Authentication Module

This module handles user registration and login using Firebase Authentication. Users can create an account using email and password or sign in using Google Sign-In. The module ensures secure authentication and session management. Once the user logs in successfully, they are redirected to the dashboard.

3.2 Profile Management Module

This module allows users to create and manage their personal profile. Users can upload a profile photo and enter personal details such as name, college, graduation year, and phone number. The profile information is stored in the database and can be updated by the user.

3.3 Resume Upload and Analysis Module

This module allows users to upload their resume in PDF format. The system analyzes the resume using Artificial Intelligence to extract skills, experience level, and important keywords. Based on the resume analysis, the system suggests suitable job roles for interview practice.

3.4 Role Selection Module

This module allows users to select the job role they want to practice for interviews. The system also shows recommended roles based on resume analysis. Users can select recommended roles or add custom roles and save them for interview practice.

3.5 Dashboard Module

The dashboard acts as the main home screen of the application. It displays important

information such as total interview sessions, average score, best score, and recent session activity. The dashboard also provides quick navigation to start interview sessions, view progress, upload resume, and manage profile.

3.6 Session Setup Module

This module allows users to configure interview settings before starting the interview. Users can select job role, difficulty level, interview category (technical, behavioral, mixed), number of questions, and choose between practice mode or mock interview mode.

3.7 Face Verification Module

This module verifies the user before starting the mock interview. The system uses face detection to ensure that only one face is present and the user is properly positioned. The mock interview starts only after successful face verification. The current version supports face presence verification, and future enhancements can include face matching using FaceNet.

3.8 Question Generation Module

This module generates interview questions using Artificial Intelligence based on the selected job role, difficulty level, and interview category. The system ensures that questions are not repeated and provides fallback questions if AI fails to generate questions.

3.9 Voice Answer Module

This module allows users to answer interview questions using voice recording. The system records the user's voice, converts speech to text, and displays the transcribed answer before submitting it for evaluation.

3.10 AI Feedback Module

This module evaluates the user's answers using Artificial Intelligence and generates scores and feedback. The feedback includes strengths, improvement suggestions, and model answers to help the user improve performance.

3.11 Mock Interview Monitoring Module

This module monitors the user during mock interviews using the device camera. The system detects if no face or multiple faces are present and shows warnings. It also tracks malpractice

count during the mock interview session.

3.12 Session Summary Module

This module displays the final result after the interview is completed. It shows overall score, question-wise scores, AI-generated summary, and performance feedback. Users can navigate to dashboard or progress page from this screen.

3.13 Session Details Module

This module stores and displays complete details of each interview session. It shows session role, date, questions, user answers, individual scores, and feedback for each question.

3.14 Progress Tracking Module

This module shows the user's performance history and analytics. It displays previous sessions, average score, best score, streak, score trends, and role-wise performance. Users can also open session details from this module.

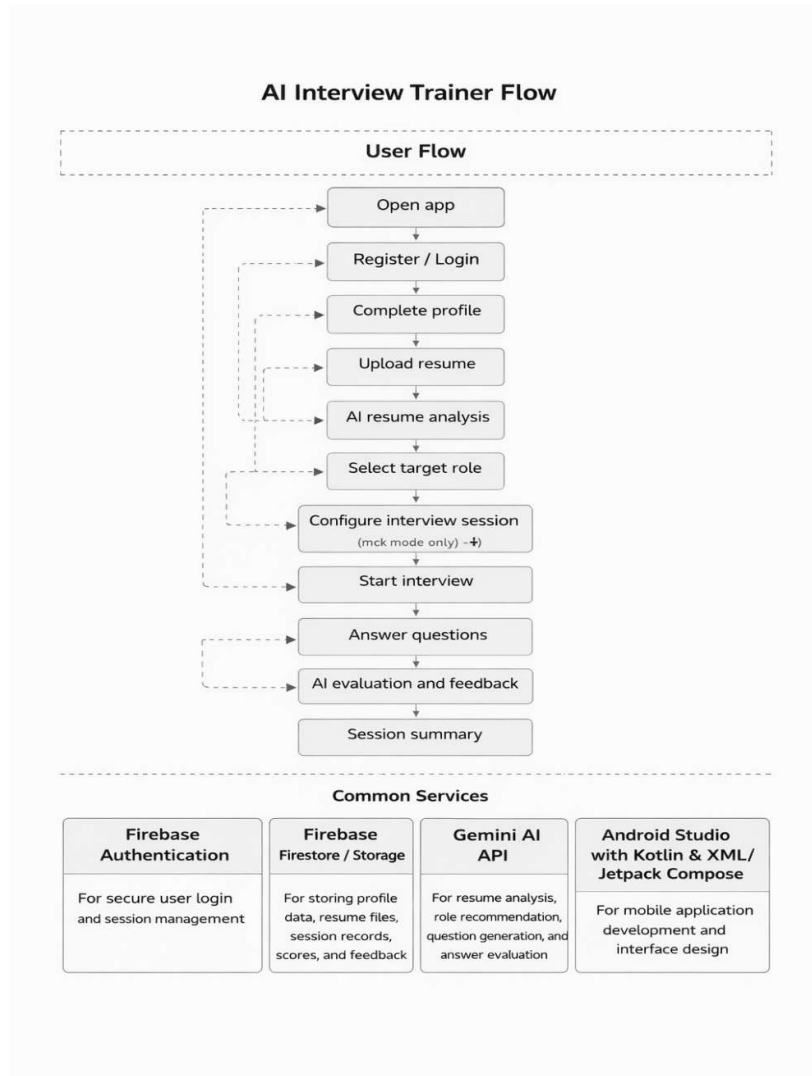
3.15 Data Storage Module

This module manages all application data storage. It stores user profile data, resume path, selected roles, interview sessions, question attempts, scores, and feedback using Firebase Firestore and local storage.

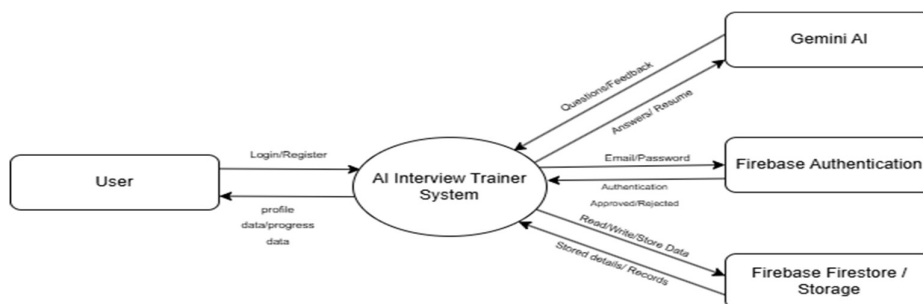
3.16 AI Integration Module

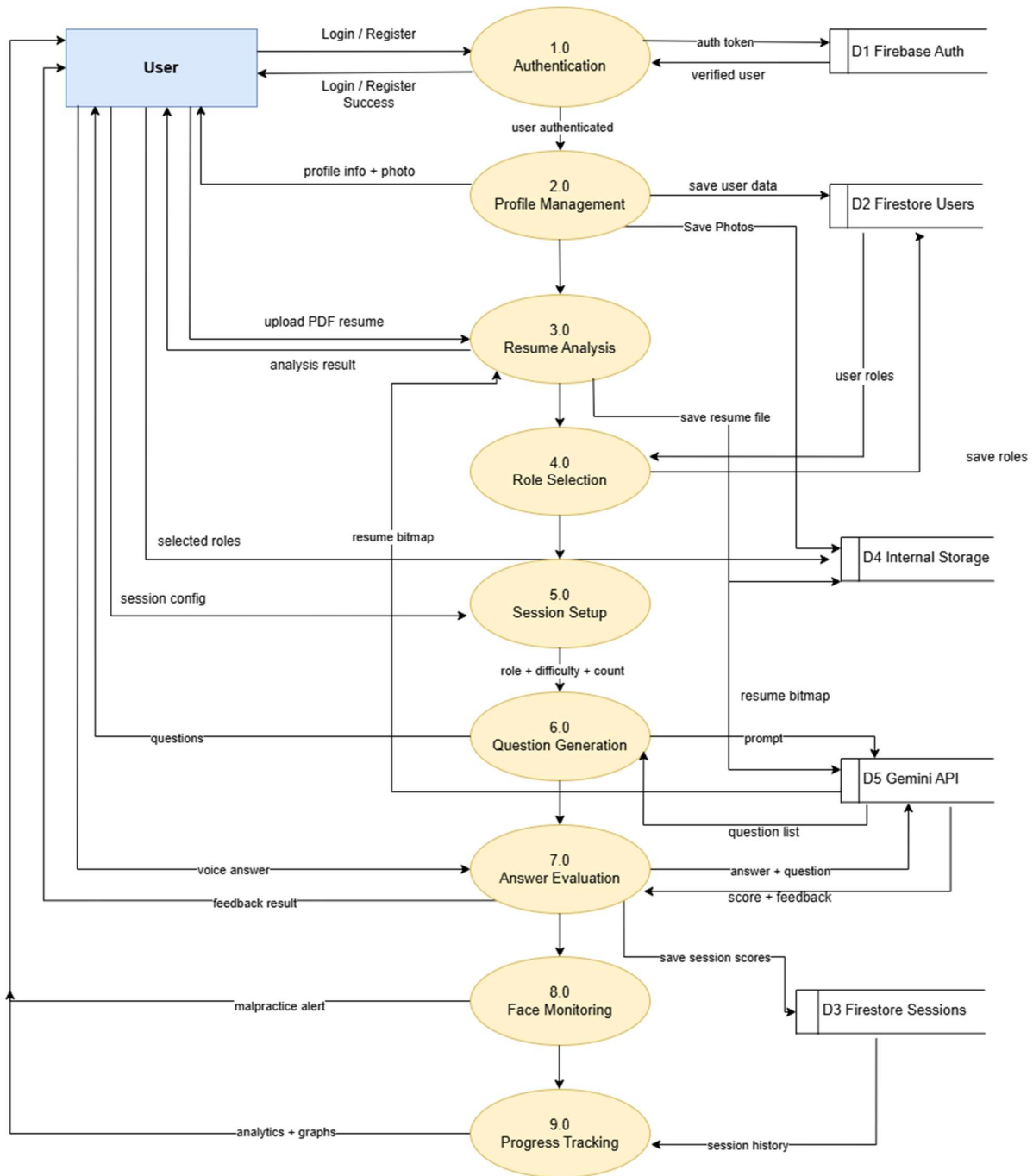
This module connects the application with Gemini AI for various AI functionalities such as resume analysis, role recommendation, question generation, answer evaluation, and session summary generation.

3.7 DATA FLOW DIAGRAM

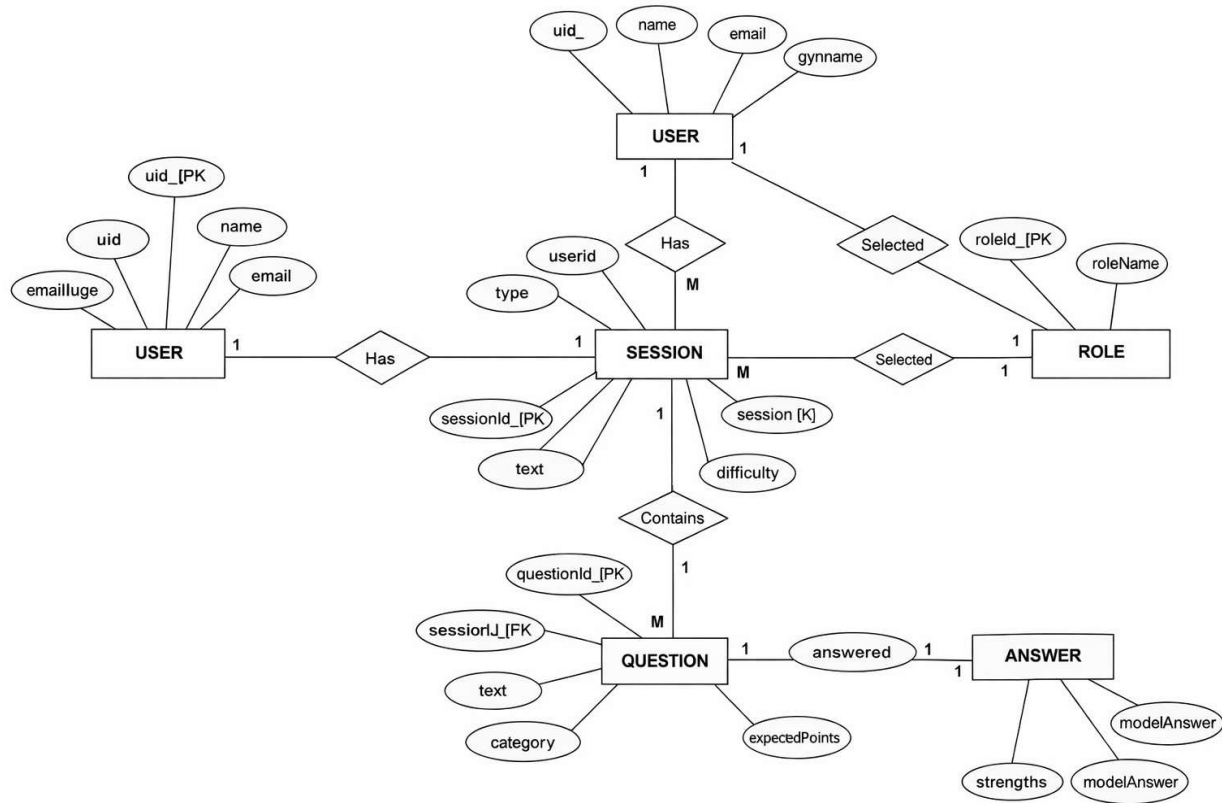


3.7.1 LEVEL – 0 (DFD) DIAGRAM



LEVEL – 1 (DFD) DIAGRAM

4. ER DIAGRAM



ER DIAGRAM – AI INTERVIEW TRAINER

5. TECHNOLOGIES USED

The following technologies, tools, and frameworks were used in the development of the **AI Interview Trainer**:

Programming Language: Kotlin

Development Platform: Android Studio

Database: Firebase Firestore (Cloud NoSQL Database)

Other Tools: Firebase Authentication, Google ML Kit Face Detection, Gemini AI API, Material Design Components, Gradle

Kotlin

Kotlin is a modern, statically typed programming language used for Android application development. It is fully supported by Google and provides concise syntax, null safety, and improved performance compared to Java. In the AI Interview Trainer application, Kotlin is used to implement application logic, user interface interactions, data handling, and integration with Firebase and AI services.

Android Studio

Android Studio is the official Integrated Development Environment (IDE) for Android application development provided by Google. It provides tools for designing layouts, writing code, debugging, testing, and running applications on emulators or physical devices. Android Studio was used to design user interfaces using XML, implement Kotlin code, manage dependencies using Gradle, and build and deploy the application.

Firebase Firestore

Firebase Firestore is a cloud-based NoSQL database used to store and retrieve application data in real time. Firestore stores user profile data, resume details, selected roles, interview session records, scores, and feedback. Firestore provides real-time synchronization and offline support, making it suitable for mobile applications.

Firebase Authentication

Firebase Authentication provides secure user authentication using email and password login as well as Google Sign-In. It manages user sessions securely and ensures that only authenticated users can access the application.

Firebase Storage

Firebase Storage is used to store user-uploaded resume files and profile images securely in the cloud. The application retrieves resume files from storage for analysis and stores profile images for user profiles.

Google ML Kit Face Detection

Google ML Kit Face Detection is used to detect the user's face during mock interview sessions. It ensures that only one face is visible during the mock interview and helps simulate a real interview environment.

Gemini AI API

Gemini AI is used to implement Artificial Intelligence features in the application such as resume analysis, role recommendation, interview question generation, answer evaluation, and session summary generation. It provides intelligent feedback and scoring based on user answers.

Material Design Components

Material Design Components are used to design the user interface of the application. Components such as CardView, Buttons, TextInputLayout, Navigation Bar, and Progress Indicators are used to create a modern and user-friendly interface.

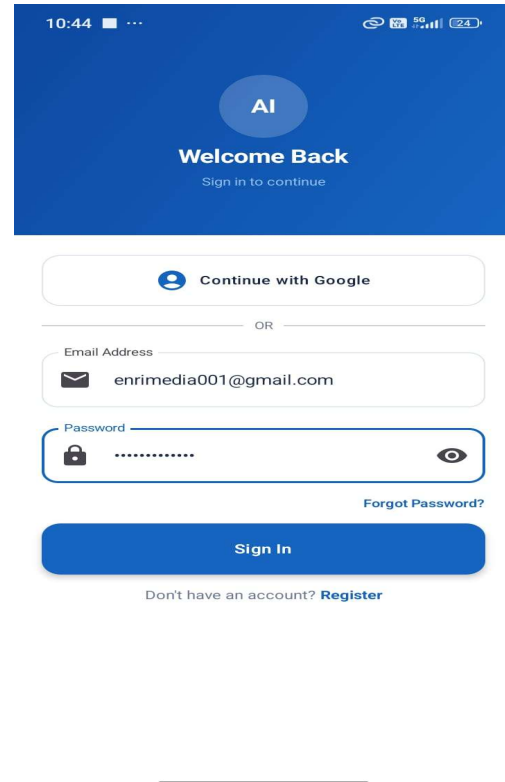
Gradle (Build Automation)

Gradle is used as the build automation tool for the project. It manages project dependencies, compiles source code, and builds the Android application into an APK file for installation.

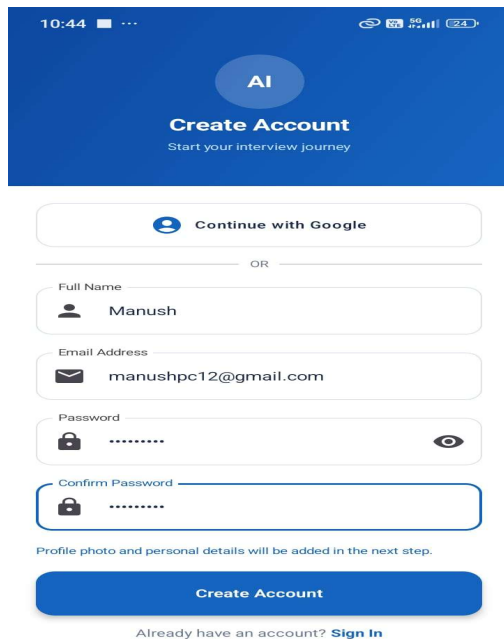
6. SCREENSHOTS



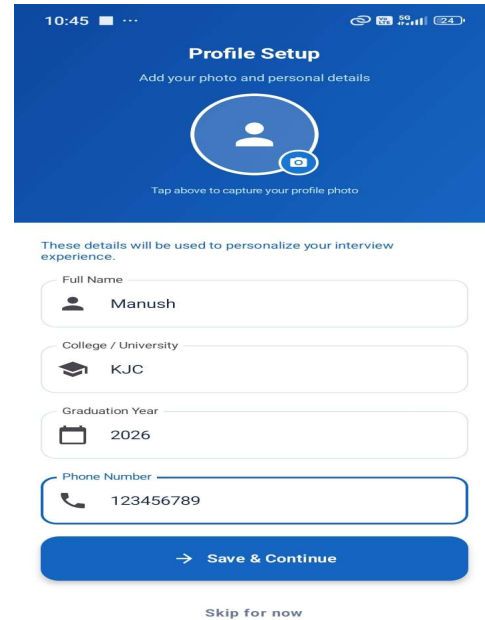
Splash Screen



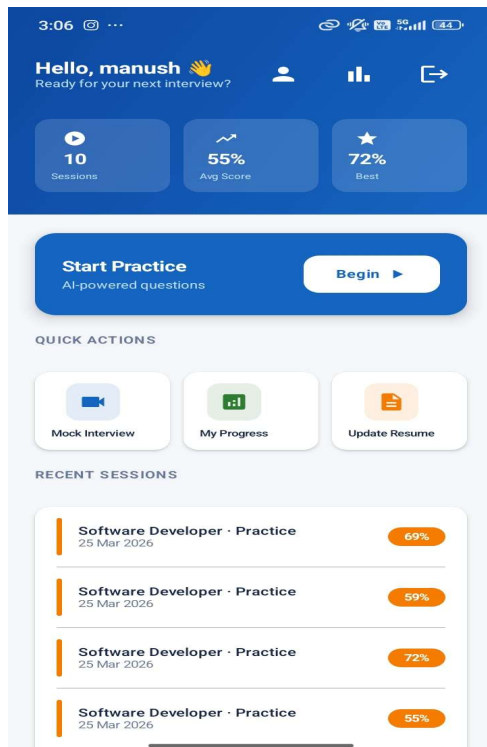
Login Screen



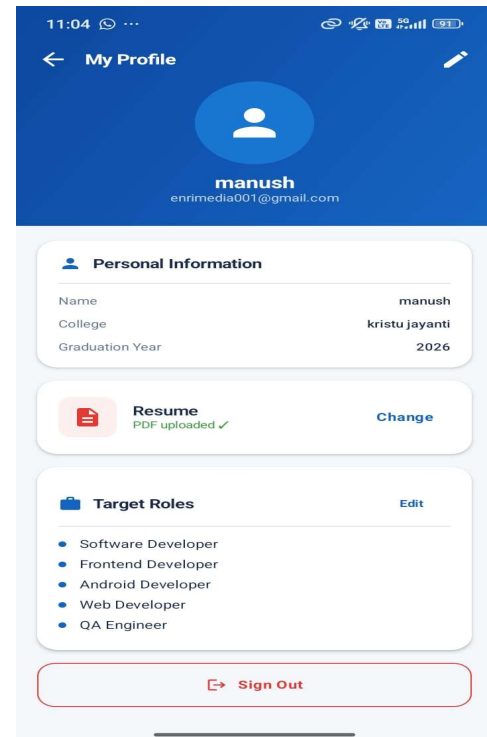
Register Page



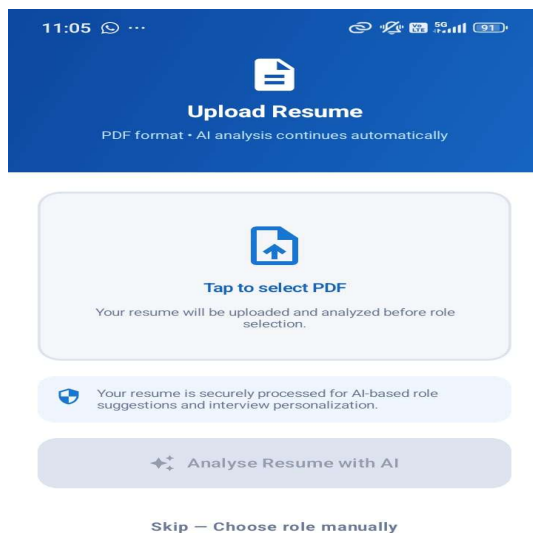
Profile Setup Page



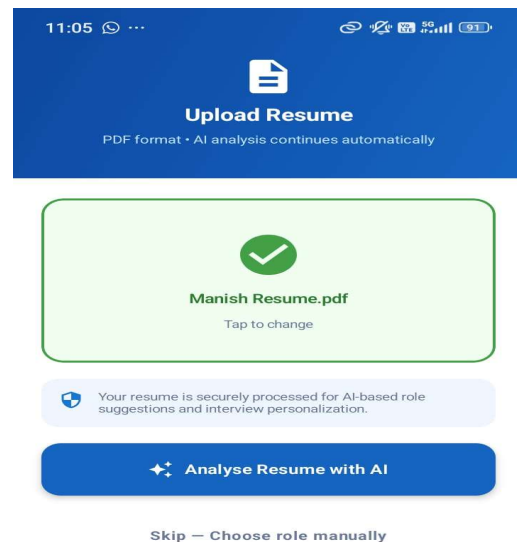
Dashboard



Profile Page



Resume Upload Page



Upload Resume-Analysis

11:05 ...

AI Analysis
Gemini is analyzing your resume

Profile Summary
AI resume analysis is temporarily unavailable. Default role suggestions are shown so you can continue.
Level: Fresher

DETECTED SKILLS
Communication Problem Solving

AI-SUGGESTED ROLES

1	Software Developer	✓
2	Frontend Developer	✓
3	Android Developer	✓
4	Web Developer	✓
5	QA Engineer	✓

→ Continue to Role Selection

Role Selection – AI Suggestions

11:06 ...

Choose Your Target Roles
AI-suggested roles and manual selection
5 selected

AI Suggested / Current Selected Roles
Software Developer, Frontend Developer, Android Developer, Web Developer, QA Engineer

+ Add custom role Add

POPULAR ROLES

✓ Software Developer ✓ Frontend Developer

Backend Developer Full Stack Developer

✓ Android Developer iOS Developer

Data Scientist Data Analyst ML Engineer

DevOps Engineer Cloud Engineer

✓ QA Engineer Product Manager

UI/UX Designer Business Analyst

Role Selection – AI Suggestions

11:06 ... Add custom role Add

POPULAR ROLES

✓ Software Developer ✓ Frontend Developer

Backend Developer Full Stack Developer

✓ Android Developer iOS Developer

Data Scientist Data Analyst ML Engineer

DevOps Engineer Cloud Engineer

✓ QA Engineer Product Manager

UI/UX Designer Business Analyst

System Administrator Cybersecurity Analyst

Database Administrator

ADDED BY YOU

× Web Developer

Save & Go to Dashboard

Role Selection – Manual

11:03 ... Session Setup

TARGET ROLE
Software Developer

DIFFICULTY LEVEL
Beginner Intermediate Advanced

INTERVIEW CATEGORY
Technical Behavioral Mixed

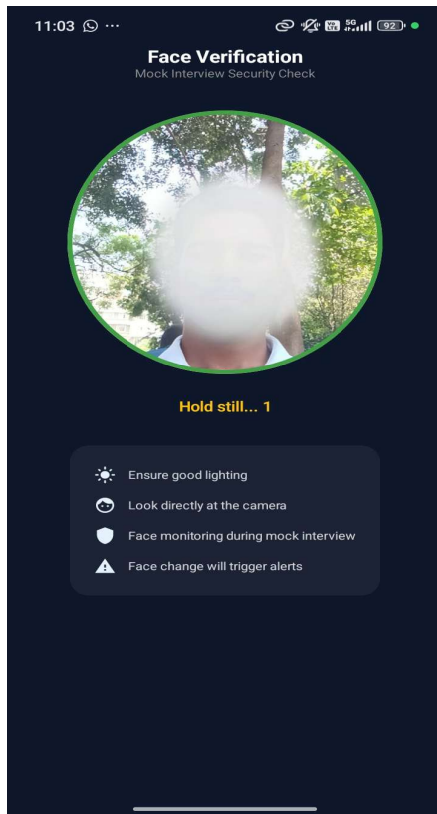
NUMBER OF QUESTIONS
5 10 15

TIME PER QUESTION
60s 90s 120s

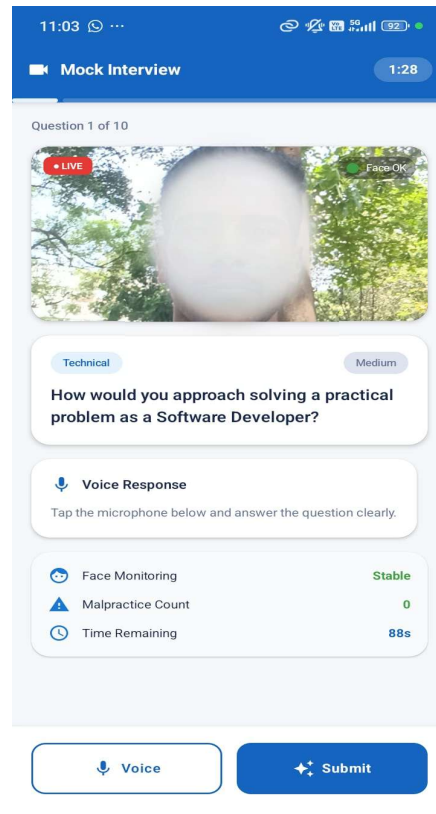
Mock Interview Mode
Face verification will be active

Start Session

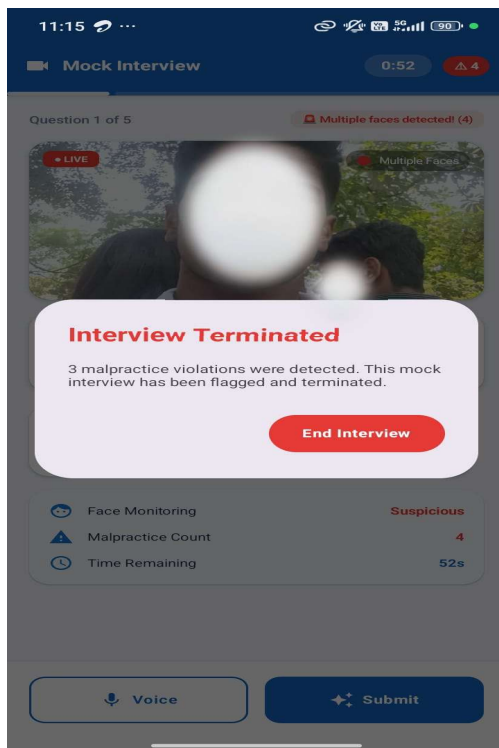
Session Setup



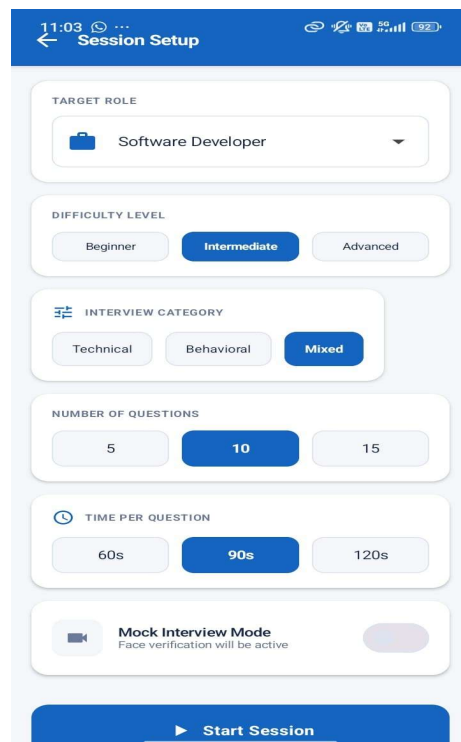
Face Verification



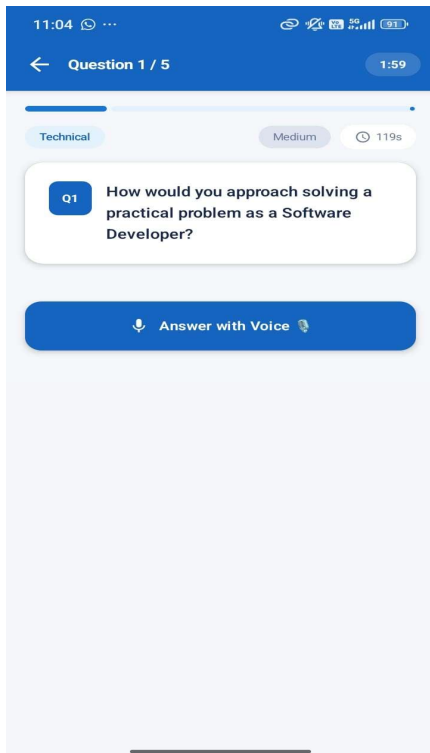
Mock Interview



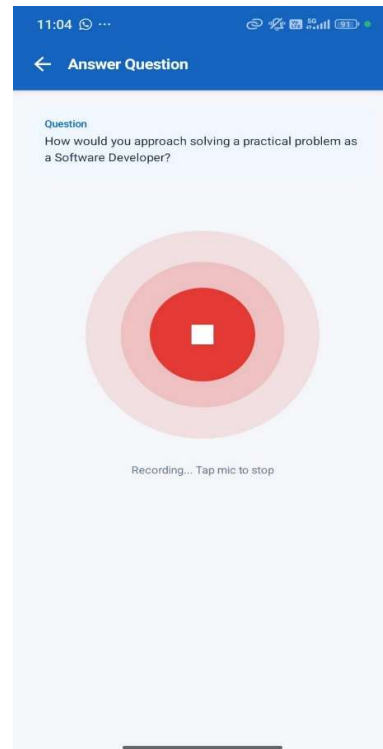
Malpractice Diction



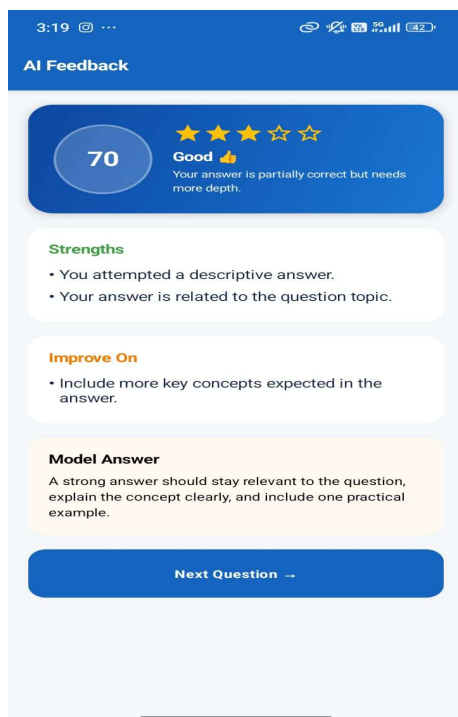
Voice Interview Session Setup



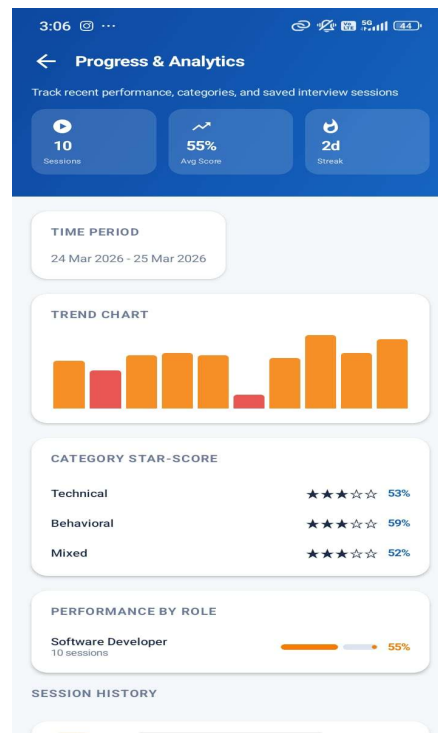
Voice Interview



Voice Interview-Recording



AI Feedback



Voice Interview-Recording

6. PROJECT FEATURES

The AI Interview Trainer includes the following key features and functionalities:

6.1 User Authentication (Register & Login)

- Users can create an account using email and password.
- Users can log in using email/password or Google Sign-In.
- Secure authentication is implemented using Firebase Authentication.
- User session is maintained across app launches unless the user logs out.
- User account details are securely stored in Firebase Firestore.

6.2 Profile Management

- Users can create and update their personal profile.
- Users can upload a profile photo.
- Users can enter personal details such as name, college, graduation year, and phone number.
- Profile information is stored securely and can be updated anytime.

6.3 Resume Upload and AI Analysis

- Users can upload their resume in PDF format.
- Resume is stored securely in Firebase Storage.
- The system analyzes the resume using AI to extract skills and experience level.
- The system suggests suitable job roles based on resume analysis.

6.4 Role Selection

- Users can select recommended job roles suggested by AI.
- Users can add custom job roles.
- Users can save selected roles for interview practice.

6.5 Dashboard

- Dashboard displays total interview sessions attended by the user.
- Dashboard displays average score and best score.
- Dashboard shows recent interview sessions.
- Dashboard provides quick access to interview, progress, resume, and profile modules.

6.6 Interview Session Setup

- Users can select job role before starting the interview.
- Users can select difficulty level (Easy, Medium, Hard).
- Users can select interview category (Technical, Behavioral, Mixed).
- Users can select number of questions.
- Users can choose between Practice Mode and Mock Interview Mode.

6.7 Face Verification and Monitoring

- The system detects the user's face before starting mock interview.
- The system ensures only one face is visible during mock interview.
- The system shows warnings if no face or multiple faces are detected.
- The system tracks malpractice during mock interview sessions.

6.8 AI Question Generation

- The system generates interview questions using AI.
- Questions are generated based on selected role and difficulty level.
- Questions can be technical, behavioral, or mixed.
- The system avoids repeated questions.
- Fallback questions are provided if AI fails to generate questions.

6.9 Voice Answer Recording

- Users can answer interview questions using voice recording.
- The system converts speech to text.
- The transcribed answer is shown to the user before submission.
- The answer is submitted for AI evaluation.

6.10 AI Feedback and Evaluation

- The system evaluates answers using AI.
- The system provides score for each answer.
- The system shows strengths and improvement suggestions.
- The system shows model answers for reference.
- The system provides detailed feedback after the interview.

6.11 Session Summary

- The system displays overall interview score.
- The system displays question-wise scores.
- The system displays AI-generated summary and feedback.
- Users can navigate to dashboard or progress page after session.

6.12 Progress Tracking

- Users can view previous interview sessions.
- The system displays average score and best score.
- The system shows performance trends and streaks.
- The system shows role-wise performance.
- Users can open detailed session records.

6.13 Data Storage and Security

- User data is stored securely in Firebase Firestore.
- Resume files are stored in Firebase Storage.
- Interview session data, scores, and feedback are stored in database.
- Secure authentication ensures data privacy and security.

6.14 Clean and User-Friendly Interface

- The application has a simple and modern user interface.
- Material Design components are used for UI design.
- The application is responsive and works on different Android screen sizes.
- The interface is designed to be easy to use for students and job seekers.

7. TESTING

The following test cases were used to verify the functionality of the AI Interview Trainer application.

Test Case ID	Module	Test Scenario	Input	Expected Result	Actual Result	Status
TC01	Authentication Module	User registration with valid details	Valid email, password, user details	User account should be created successfully	Account created successfully	Pass
TC02	Authentication Module	User login with valid credentials	Registered email and password	User should login and navigate to dashboard	Login successful	Pass
TC03	Authentication Module	Google Sign-In	Valid Google account	User should login successfully using Google account	Google login successful	Pass
TC04	Profile Management Module	Update profile details	Name, college, year, phone, photo	Profile data should be saved and displayed	Profile updated successfully	Pass
TC05	Resume Upload and Analysis Module	Upload resume PDF	Valid PDF file	Resume should be uploaded and stored successfully	Resume uploaded successfully	Pass

Test Case ID	Module	Test Scenario	Input	Expected Result	Actual Result	Status
TC06	Resume Upload Analysis Module	and AI analysis of resume	Uploaded resume	Skills, experience level, and recommended roles should be generated	Resume analyzed successfully	Pass
TC07	Role Selection Module	Select recommended role	AI suggested role	Selected role should be saved	Role saved successfully	Pass
TC08	Role Selection Module	Add custom role	Custom role name	Custom role should be added and saved	Custom role added successfully	Pass
TC09	Dashboard Module	Display dashboard stats	User session data	Dashboard should show total sessions, average score, and best score	Dashboard displayed correctly	Pass
TC10	Session Setup Module	Configure interview session	Role, difficulty, category, question count, mode	Session should start with selected configuration	Session started successfully	Pass
TC11	Face Verification	Detect single face before mock	User face in camera	System should allow interview	Face verified	Pass

Test Case ID	Module	Test Scenario	Input	Expected Result	Actual Result	Status
	Module	interview		to start	successfully	
TC12	Face Verification Module	Detect multiple faces	More than one face in camera	System should show warning and block interview start	Warning displayed correctly	Pass
TC13	Question Generation Module	Generate interview questions	Selected role, difficulty, category	AI should generate questions	Questions generated successfully	Pass
TC14	Voice Answer Module	Record voice answer	Spoken answer	Voice should be recorded and converted to text	Voice transcribed successfully	Pass
TC15	AI Feedback Module	Evaluate answer using AI	User answer text	Score, strengths, improvements, and model answer should be displayed	Feedback generated successfully	Pass
TC16	Mock Interview Monitoring Module	Detect no face during mock interview	Camera with no face	System should show warning	Warning displayed successfully	Pass
TC17	Session Summary	Show interview	Completed interview	Overall score, question-wise	Summary displayed	Pass

Test Case ID	Module	Test Scenario	Input	Expected Result	Actual Result	Status
	Module	summary	session	scores, and summary should be displayed	correctly	
TC18	Session Details Module	Open previous session details	Selected session	Detailed record of session should be shown	Session details displayed correctly	Pass
TC19	Progress Tracking Module	View progress history	Previous session data	Average score, best score, and trends should be shown	Progress displayed correctly	Pass
TC20	Data Storage Module	Store session and feedback data	Completed interview session	Session data should be saved in Firestore	Data stored successfully	Pass

8. USER GUIDE

This section provides instructions on how to install and use the AI Interview Trainer application.

Installation Instructions :

I) Installing from APK

Step 1: Download the APK file to your Android mobile device.

Step 2: Open your device's File Manager and navigate to the downloaded APK file.

Step 3: Tap on the APK file and select Install.

Step 4: If a pop-up appears saying installation is blocked for unknown sources, tap Settings and enable Allow from this source.

Step 5: Tap the Install button to proceed.

Step 6: Once installed, open the AI Interview Trainer application from your app drawer.

II) Installing via USB Debugging (Android Studio)

Step 1: Open the AI Interview Trainer project in Android Studio.

Step 2: Connect your Android phone to the computer using a USB cable.

Step 3: If Developer Mode is not enabled, go to Settings → About Phone → Tap Build Number seven times.

Step 4: Go to Settings → Developer Options → Enable USB Debugging.

Step 5: Select your connected device from the device dropdown in Android Studio.

Step 6: Click the Run button in Android Studio to build and install the application on your device.

III) Installing via Wireless Debugging

Step 1: Ensure Developer Mode is enabled on your phone.

Step 2: Go to Settings → Developer Options → Wireless Debugging and enable it.

Step 3: Ensure your phone and computer are connected to the same Wi-Fi network.

Step 4: Open Android Studio and navigate to Device Manager.

Step 5: Select Pair Device using QR Code or Pairing Code from Android Studio and match

the selection on your phone.

Step 6: After pairing is successful, click the Run button in Android Studio to install the application.

User Interface Guide

1. Registration / Login

When the application is opened for the first time, the user is directed to the Register screen to create an account using email and password or Google Sign-In. After successful registration, the user can log in using their credentials. The application maintains the user session until the user logs out.

2. Profile Setup

After login, the user is asked to complete their profile by entering details such as name, college, graduation year, phone number, and profile photo. This information is stored and used for interview session records and performance tracking.

3. Resume Upload and Analysis

The user can upload their resume in PDF format. The system analyzes the resume using Artificial Intelligence and extracts skills, experience level, and suggests suitable job roles based on the resume.

4. Role Selection

The system displays recommended job roles based on resume analysis. The user can select recommended roles or add custom roles for interview practice.

5. Dashboard (Home Screen)

After login, the user is redirected to the Dashboard. The dashboard displays total interview sessions, average score, best score, and recent session activity. The dashboard provides quick access to start interview, view progress, upload resume, and manage profile.

6. Interview Session Setup

Before starting the interview, the user selects job role, difficulty level, interview category, number of questions, and interview mode (Practice Mode or Mock Interview Mode). After selecting the options, the user starts the interview session.

7. Mock Interview and Face Verification

If mock interview mode is selected, the system verifies the user's face before starting the interview. The system monitors the user during the interview and shows warnings if no face or multiple faces are detected.

8. Answering Interview Questions

The system displays interview questions one by one. The user answers the questions using voice recording. The system converts speech to text and submits the answer for AI evaluation.

9. AI Feedback and Session Summary

After the interview is completed, the system displays the overall score, question-wise scores, strengths, improvement suggestions, and model answers. The system also generates an AI summary of the interview performance.

10. Progress Tracking

The user can view previous interview sessions, average score, best score, and performance trends in the progress section. The user can also open detailed session records to review answers and feedback.

9 . CONCLUSION

The AI Interview Trainer is a fully functional Android-based application developed to help students and job seekers prepare for interviews using Artificial Intelligence. The application integrates AI technologies, Firebase cloud services, and Android development tools to create an intelligent interview preparation platform. The system provides features such as resume analysis, role recommendation, AI-generated interview questions, voice-based answer evaluation, mock interview monitoring, and performance tracking.

The development of the AI Interview Trainer provided valuable practical experience in Android application development using Kotlin, working with Firebase Authentication and Firebase Firestore, integrating Artificial Intelligence services, and implementing face detection using Google ML Kit. The project helped in understanding how real-world mobile applications are designed using modular architecture and cloud-based databases.

The application successfully achieves its main objectives, which include helping users practice interviews, improving communication skills, providing AI-based performance feedback, and tracking user progress over time. The resume analysis and AI question generation features help users prepare for role-specific interviews, while the mock interview monitoring feature simulates real interview conditions. The progress tracking feature allows users to monitor their performance and identify areas for improvement.

During the development of this project, several challenges were encountered such as integrating AI APIs, implementing speech-to-text conversion, managing real-time database operations, and implementing face detection during mock interviews. These challenges were overcome through research, testing, and continuous development.

In conclusion, the AI Interview Trainer is a useful and intelligent interview preparation application that demonstrates the use of Artificial Intelligence in mobile applications. The system can be further enhanced in the future by adding features such as facial expression analysis, video interview analysis, emotion detection, and personalized learning recommendations. This project demonstrates how AI can be used to improve interview preparation and help users perform better in real-world interviews.